Chapter 2: The Project Management and Information Technology Context

Information Technology Project Management, Ninth Edition

Note: See the text itself for full citations.

Twittie Senivongse (ed.)

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Learning Objectives (1 of 2)

- Define the systems view of project management and how it applies to information technology (IT) projects
- Summarize organizations, including the four frames, organizational structures, and organizational culture
- Explain why stakeholder management and top management commitment are critical for a project's success

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Learning Objectives (2 of 2)

- Distinguish between project life cycle and product life cycle
- Discuss the unique attributes and diverse nature of IT projects
- Summarize recent trends affecting IT project management, including globalization, outsourcing, virtual teams, and agile project management

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

A Systems View of Project Management

- You cannot run projects in isolation
- Projects must operate in a broad organizational environment
  - Taking a holistic view of carrying out projects within the context of the organization
- Project managers need to use systems thinking and apply systems management
  - Systems thinking views interacting components that work within an environment to fulfill some purpose
  - Systems management – addressing issues in business, technological, and organizational components of a system

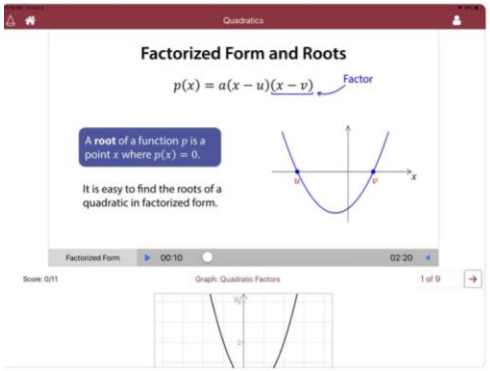
Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

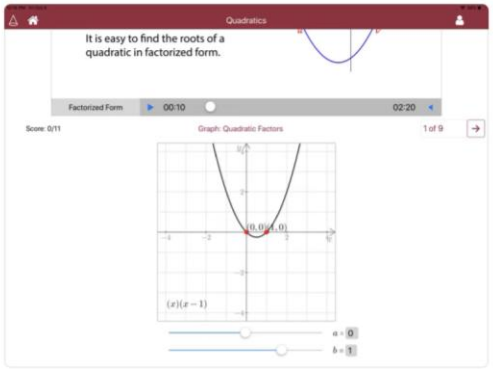
Case Study: Tablet-based interactive course materials (1 of 2)



- Tom, Director of IT at a college, wanted students to learn from interactive course materials, created by instructors, which run on tablets, just like another college that he had visited
- Benefit: Students could learn anytime, anywhere, and it could help reduce the cost of textbooks





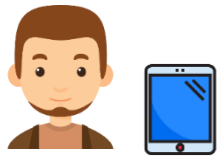




Information Technology Project Management, Ninth Edition. © 2019 Cengage. May ı part, except for use as permitted in a license distributed with a certain product or service or other use.

Icon made by Freepik from www.flaticon.com

Case Study: Tablet-based interactive course materials (2 of 2)



- Overlooking systems management, Tom and his staff developed a plan to have students lease or buy tablets at the college (if not already own one) and sent emails to describe the plan to the faculty in September but not much response.
- At a faculty meeting in February
  - Chairs of History, English, Economics:
    - College was not a technical training school
    - No time to write their own course materials
    - Students liked the book they used
    - Students could buy electronic books but preferred print version anyway
  - Chair of Computer Science
    - Students already owned more powerful laptops
  - Director of Adult Education
    - Students would complain about an increase in fees

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Three-Sphere Model for Systems Management

Organizational issues are often the most difficult part of working on and managing projects

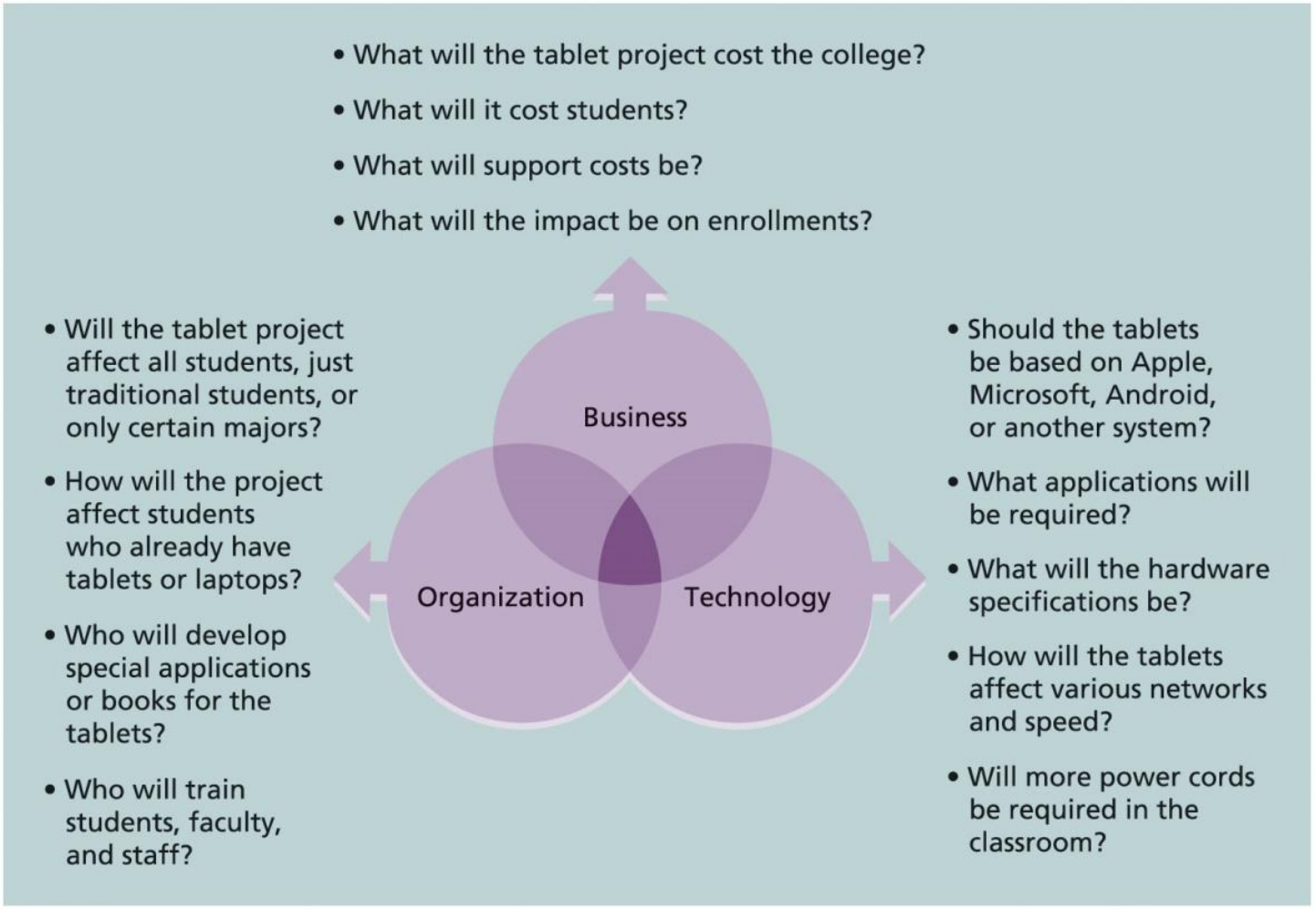


FIGURE 2-1 Three-Sphere model for systems management

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Undestanding Organizations: The Four Frames

Structural frame: Roles and responsibilities, coordination, and control. Organizational charts help describe this frame.	**Human resources frame:** Providing harmony between needs of the organization and needs of people.
Political frame: Coalitions composed of varied individuals and interest groups. Conflict and power are key issues.	**Symbolic frame:** Symbols and meanings related to events. Culture, language, traditions, and image are all parts of this frame.

Source: Bolman and Deal.

FIGURE 2-2 Perspectives on organizations[2]

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Organizational Structures

Structural frame

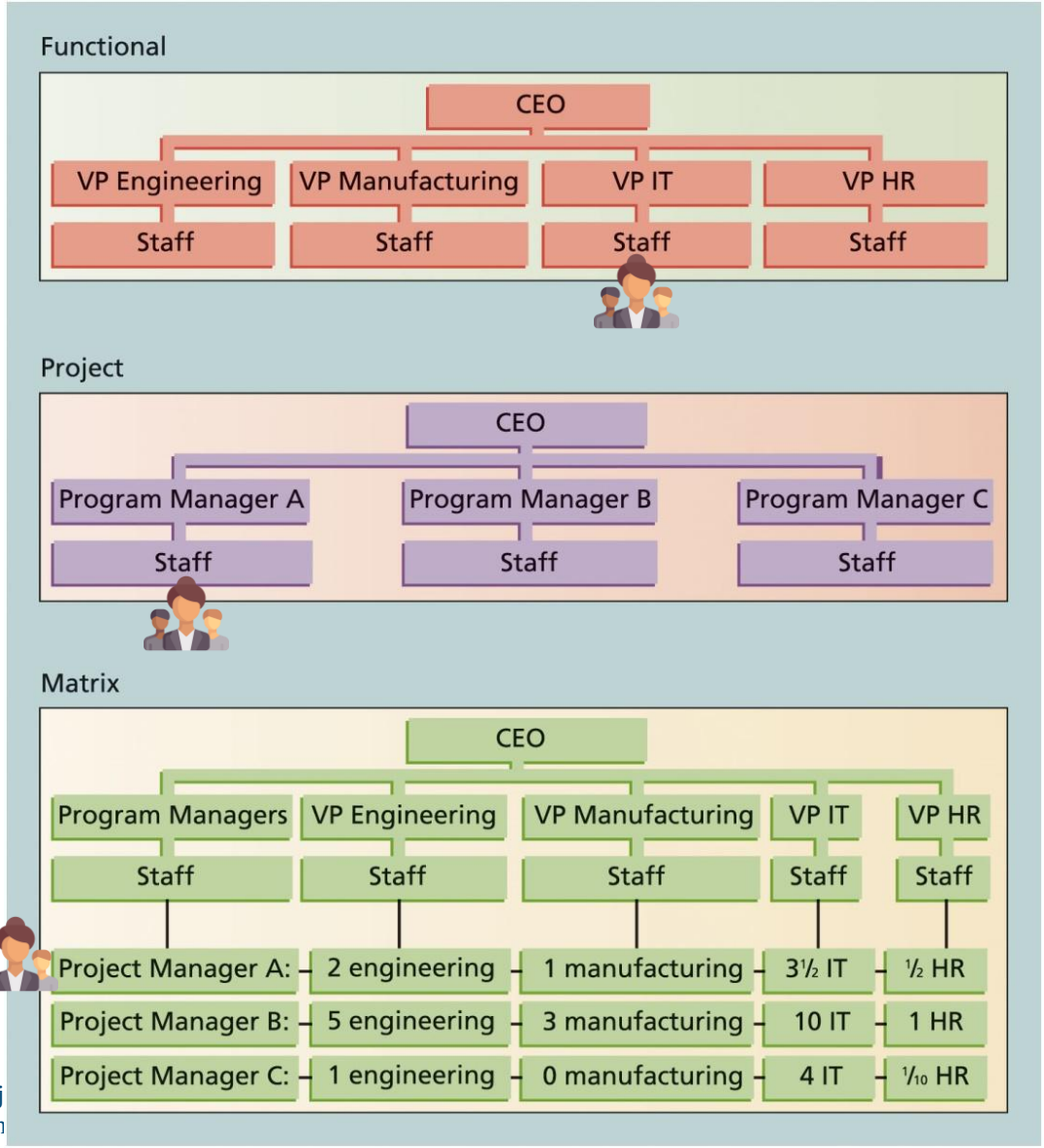


FIGURE 2-3 Functional, project, and matrix organizational structures

Information Technology Proj ole or in part, except for use as permitted in a licen ssroom use.

Organizational Culture

Symbolic frame

- Organizational culture is a set of shared assumptions, values, and behaviors that characterize the functioning of an organization
- Ten characteristics of organizational culture:
  - Member identity*
  - Group emphasis*
  - People focus
  - Unit integration*
  - Control
  - Risk tolerance*
  - Reward criteria*
  - Conflict tolerance*
  - Means-ends orientation
  - Open-systems focus*

- *Project work is most successful in an organizational culture where these items are strong/high and other items are balanced.

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Tom's mistakes regarding the four frames of organizations



Structural frame: Roles and responsibilities, coordination, and control. Organizational charts help describe this frame.

Tom could have asked for support from President or Deans first.

Human resources frame: Providing harmony between needs of the organization and needs of people.

Mismatches occurred between the needs of the organization (new interactive course materials) and the needs of people (Department Chairs did not need new course materials).

Political frame: Coalitions composed of varied individuals and interest groups. Conflict and power are key issues.

Conflict with Department Chairs in faculty meeting.

Symbolic frame: Symbols and meanings related to events. Culture, language, traditions, and image are all parts of this frame.

Students of some majors preferred textbooks. Email to faculty at the start of semester in September got no attention.

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Importance of Top Management Commitment (1 of 2)

- People in top management positions are key stakeholders in projects
- A very important factor in helping project managers successfully lead projects is the level of commitment and support they receive from top management
- Some projects have a senior manager called a project champion (or project advocate) who acts as a key proponent for a project.
  - Informal role whose main goal is to make the project succeed by addressing different obstacles while making sure that the stakeholders are always satisfied with the project
  - Project champion role may be embodied by the project sponsor

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Importance of Top Management Commitment (2 of 2)

- How top management can help project managers
  - Providing adequate resources
  - Approving unique project needs in a timely manner
  - Getting cooperation from other parts of the organization
  - Mentoring and coaching on leadership issues

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Need for Organizational Standards

- Standards and guidelines help project managers be more effective
- Senior management can encourage
  - the use of standard forms and software for project management
  - the development and use of guidelines for writing project plans or providing status information
  - the creation of a project management office (PMO) or center of excellence

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Project Life Cycle (1 of 2)

- A project life cycle is a collection of project phases (PMBOK® Guide, Sixth Edition, 2017)
  - Starting the project
  - Organizing and preparing
  - Carrying out the work
  - Finishing the project
- A project life cycle defines
  - what work will be performed in each phase
  - what deliverables will be produced and when
  - who is involved in each phase, and
  - how management will control and approve work produced in each phase
- A deliverable is a product or service produced or provided and delivered as part of a project

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Project Life Cycle (2 of 2)

- In early phases of a project life cycle
  - resource needs are usually lowest
  - the level of uncertainty (risk) is highest
  - project stakeholders have the greatest opportunity to influence the project
- In middle phases of a project life cycle
  - the certainty of completing a project improves
  - more resources are needed
- The final phase of a project life cycle focuses on
  - ensuring that project requirements were met
  - the sponsor approves completion of the project

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

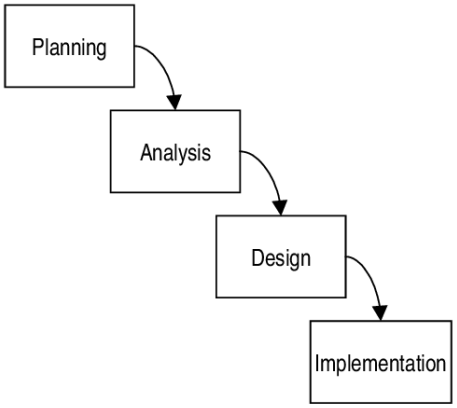
Product Life Cycles (1 of 4)

- A product life cycle is a process used to define, create, and deliver products.
- The Systems Development Life Cycle (SDLC) is a framework for describing the phases of developing information systems
- Systems development projects (PMBOK® Guide, Sixth Edition, 2017) can follow:
  - Predictive life cycle, e.g. waterfall, prototyping
    - large portion of the project is attempted to clarify requirement of the entire system and then to produce a design
    - low degree of change, low frequency of delivery
  - Iterative life cycle, e.g. spiral
    - high degree of change, low frequency of delivery
  - Incremental life cycle, e.g. phased development
    - low degree of change, high frequency of delivery
  - Adaptive life cycle (or agile)
    - high degree of change, high frequency of delivery
  - Hybrid life cycle

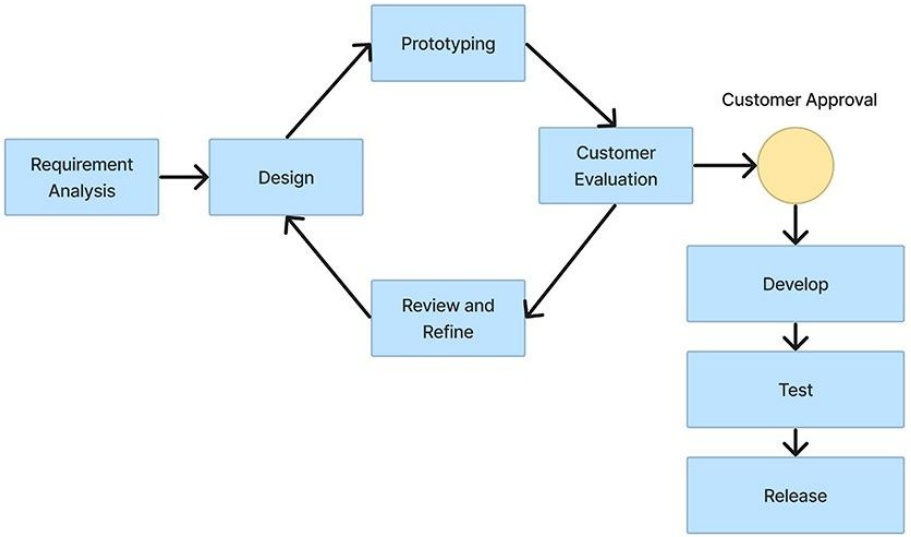
Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Product Life Cycles (2 of 4)

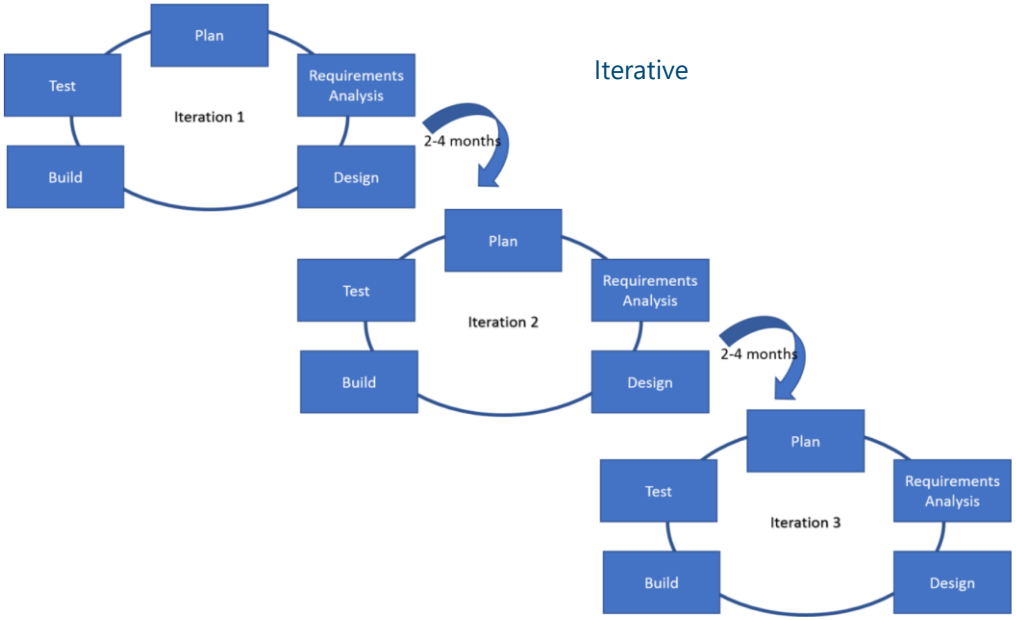
Predictive (waterfall)



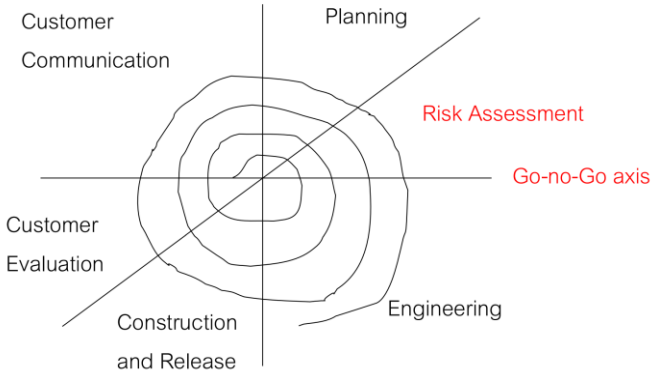
Predictive (prototyping)



Iterative



Iterative (spiral)

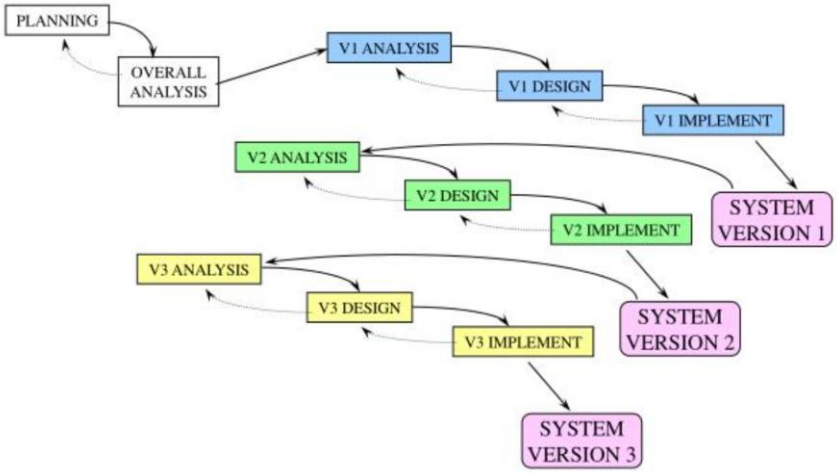


https://thedigitalattitude.com/2017/11/13/everything-you-need-to-know-about-project-life-cycles/

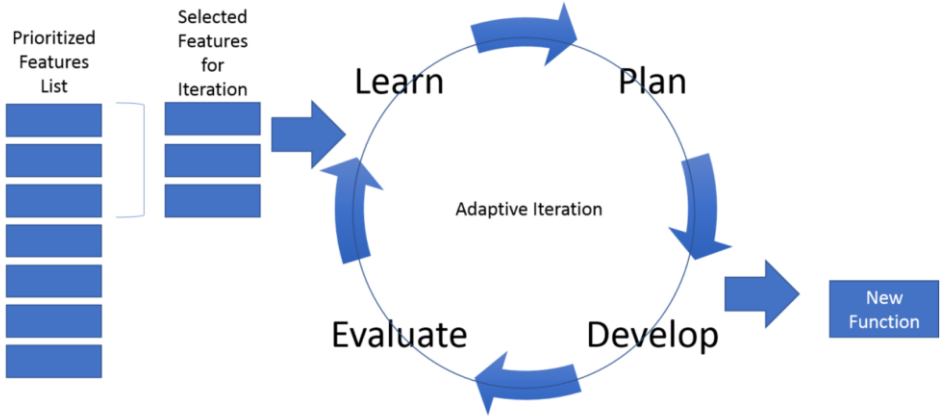
Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Product Life Cycles (3 of 4)

Incremental (phased)



Adaptive (agile)

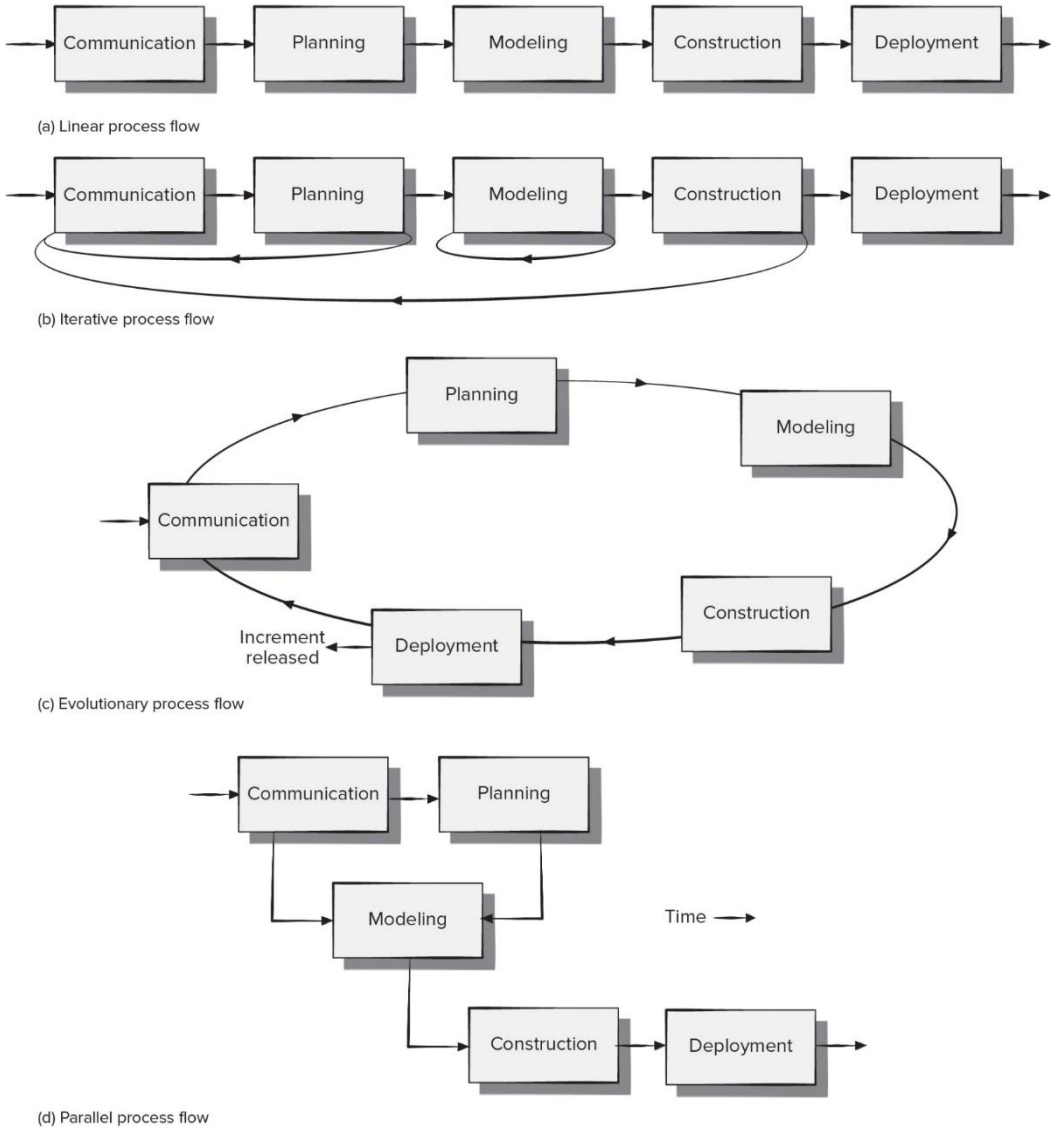


https://thedigitalattitude.com/2017/11/13/everything-you-need-to-know-about-project-life-cycles/

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Product Life Cycles (4 of 4)

Process Flow



R. S. Pressman and B. R. Maxim, Software Engineering A Practitioner's Approach, 9th edition, 2020.

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Importance of Project Phases and Management Reviews

- A project should successfully pass through each of the project phases in order to continue on to the next
- Management reviews, also called phase exits, phase gate reviews, or kill points, should occur after each phase to evaluate the project's progress, likely success, and continued compatibility with organizational goals
- It is unwise to wait until the end of project or product phases to have management inputs
  - Many projects are reviewed by management on a regular basis

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Context of IT Projects

- IT projects can be very diverse in terms of size, complexity, products produced, application areas, and resource requirements
- IT project team members often have diverse backgrounds and skill sets
  - Many companies purposely hire graduates with degrees in other fields such as business, mathematics, or liberal arts to provide different perspectives on IT projects
- IT projects use diverse technologies that change rapidly
  - Differences in technical knowledge can make communication between professionals challenging

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Recent Trends Affecting IT Project Management (1 of 3)

- Globalization: issues in global projects
  - Communications – time zones, languages, culture
  - Trust
  - Common work practices
  - Tools – communication, collaboration
- Outsourcing: organization acquires goods and/or sources from an outside source
  - Organizations may find ways to reduce cost and achieve speed and flexibility
  - Offshoring - outsourcing from another country

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Recent Trends Affecting IT Project Management (2 of 3)

- Virtual team: a group of individuals who work across time and space using communication technologies
  - Lowering costs by not requiring office space
  - Providing more expertise and flexibility by having team members across the globe work any time of day or night
  - Improving work/life balance for team members by eliminating fixed office hours and the need to travel to work
  - Increasing the potential for communications problems
  - Reducing the ability for team members to network
  - Increasing the dependence on technology to accomplish work

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

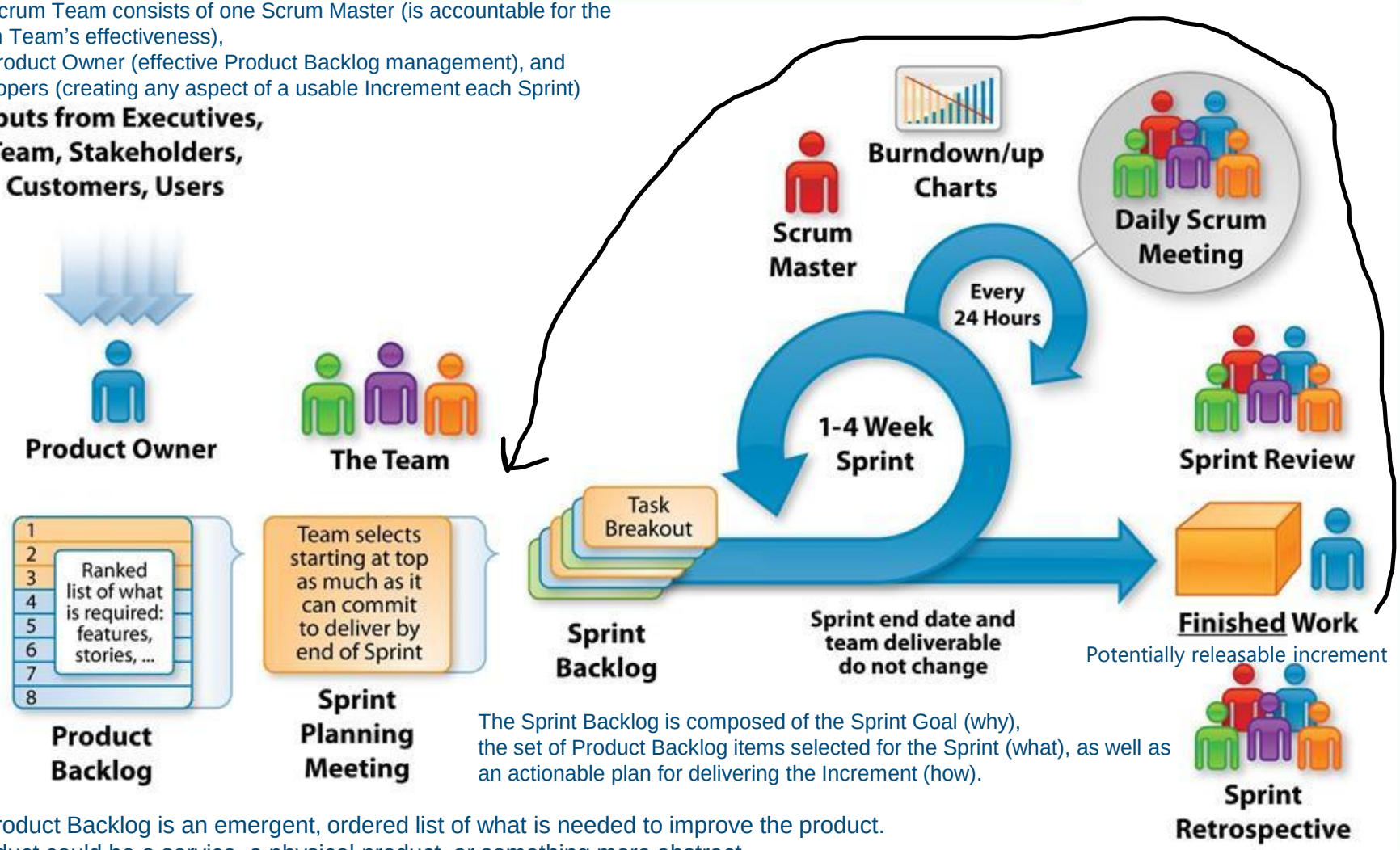
Recent Trends Affecting IT Project Management (3 of 3)

- Agile:
  - In February 2001, 17 practitioners developed manifesto for agile software development that values
    - Individuals and interactions over processes and tools
    - Working software over comprehensive documentation
    - Customer collaboration over contract negotiation
    - Responding to change over following a plan
  - Scrum is the leading agile development method
    - The term compares high-performing cross-functional teams to the scrum formation used by rugby teams

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Agile - Scrum Framework

The Scrum Team consists of one Scrum Master (is accountable for the Scrum Team's effectiveness),
one Product Owner (effective Product Backlog management), and
Developers (creating any aspect of a usable Increment each Sprint)



The Sprint Backlog is composed of the Sprint Goal (why),
the set of Product Backlog items selected for the Sprint (what), as well as
an actionable plan for delivering the Increment (how).

The Product Backlog is an emergent, ordered list of what is needed to improve the product.
A product could be a service, a physical product, or something more abstract.

Photo credit: http://edelalon.com/blog/2017/09/scrum/

http://www.scrumguides.org/scrum-guide.html

Chapter Summary

- Project managers need to take a systems approach when working on projects
- Organizations have four different frames: structural, human resources, political, and symbolic
- The structure and culture of an organization have strong implications for project managers
- Projects should successfully pass through each phase of the project life cycle
- Project managers need to consider several factors due to the unique context of information technology projects
- Recent trends affecting IT project management include globalization, outsourcing, virtual teams, and agile project management

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.